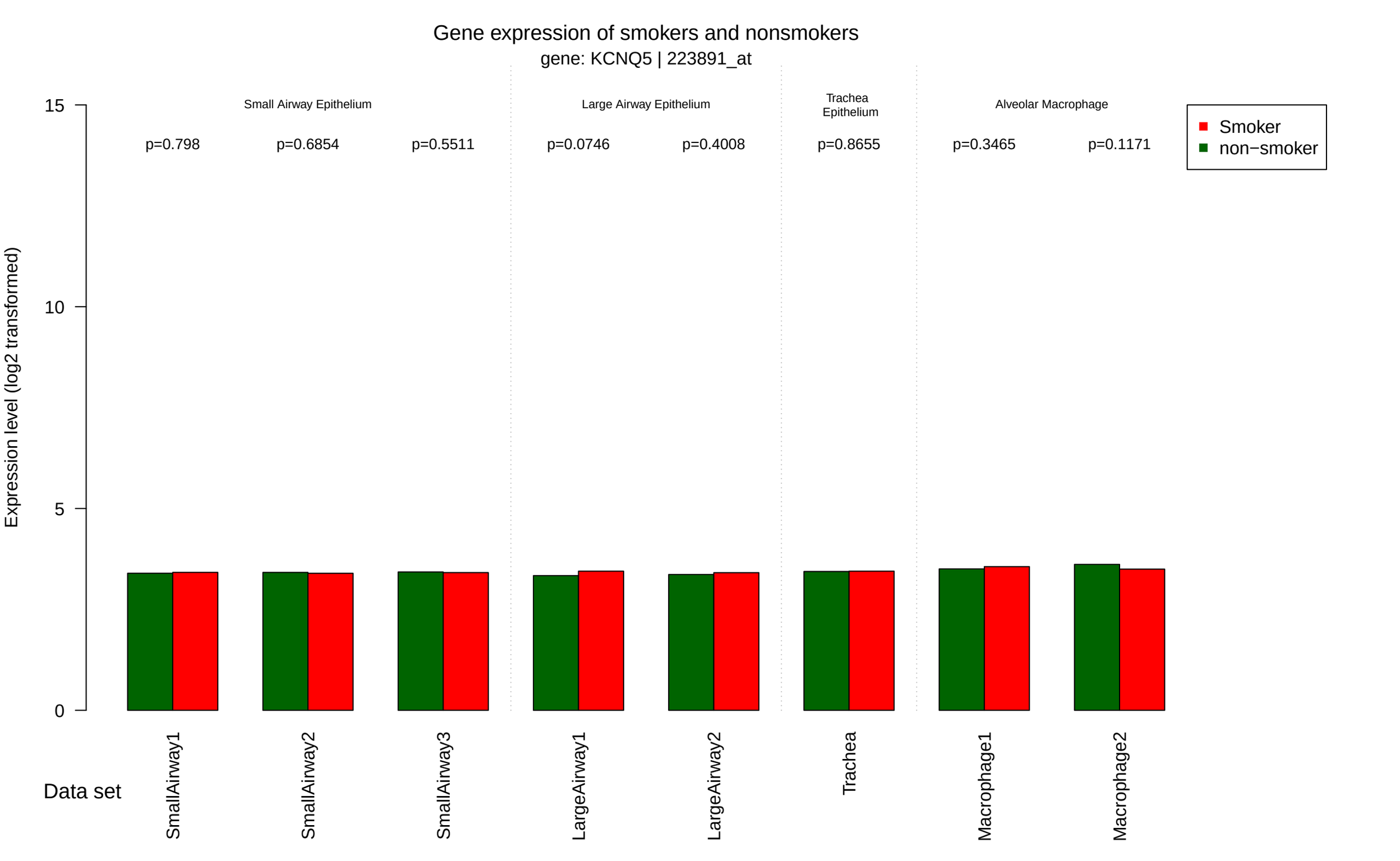
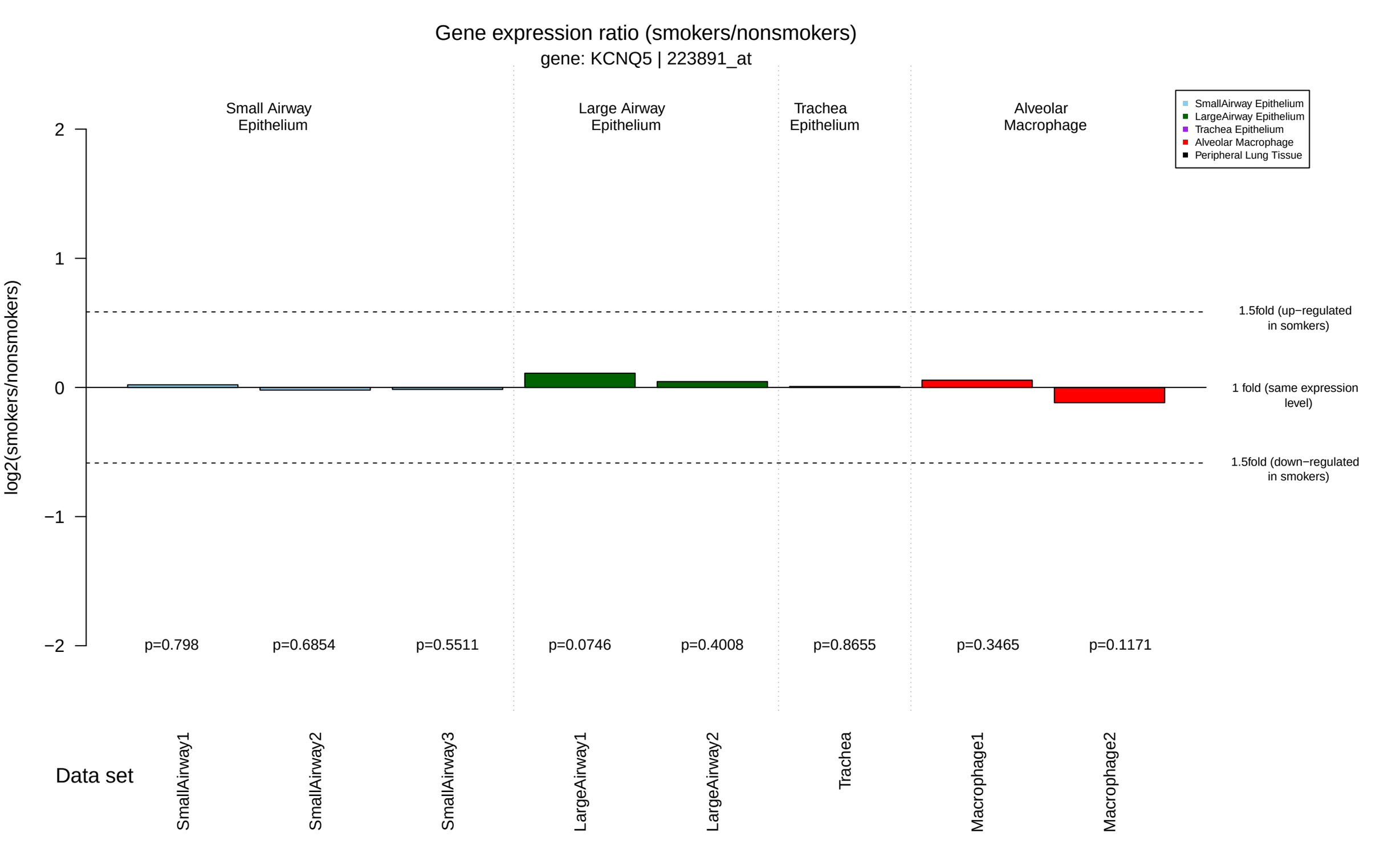
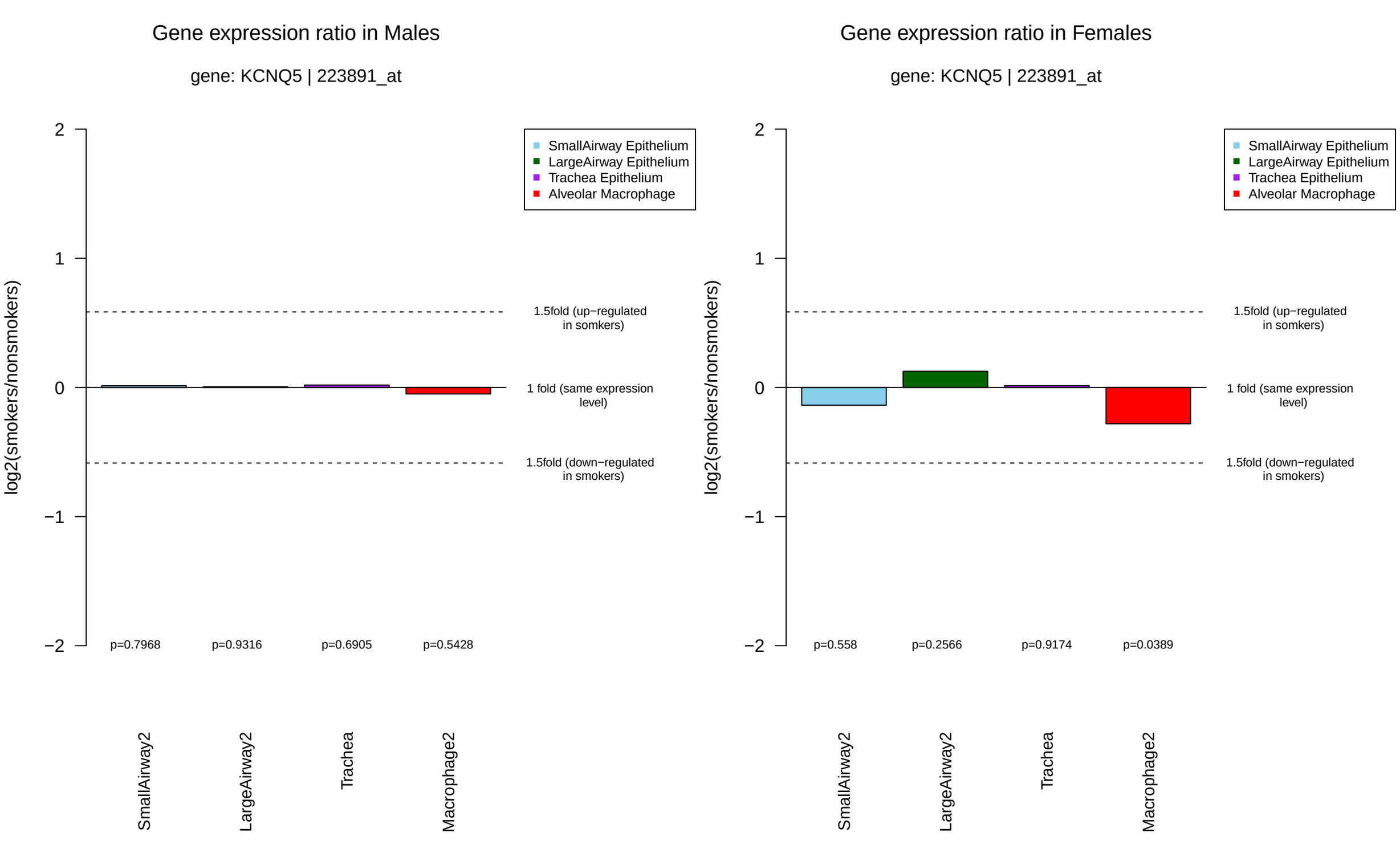
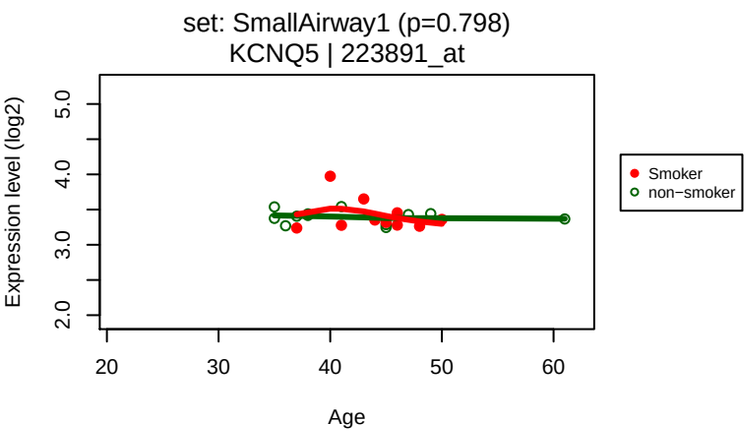
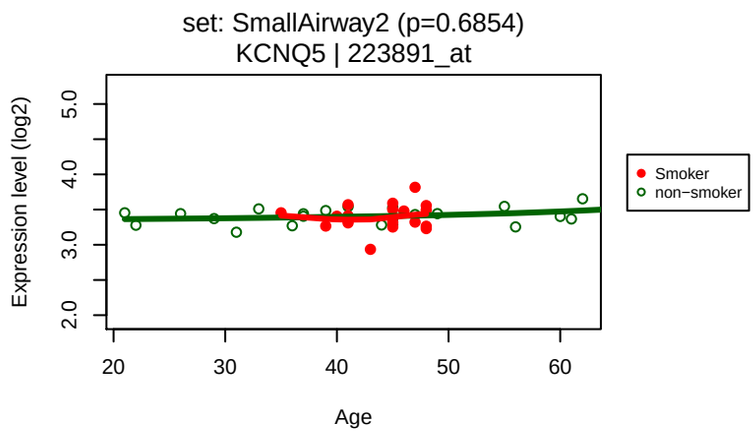
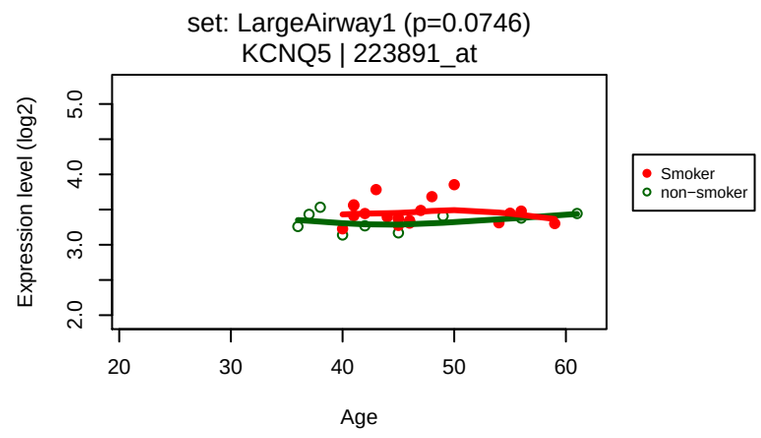

Gene expression of smokers and nonsmokers
gene: KCNQ5 | 223891_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.798
p=0.6854
p=0.5511
p=0.0746
p=0.4008
p=0.8655
p=0.3465
p=0.1171
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

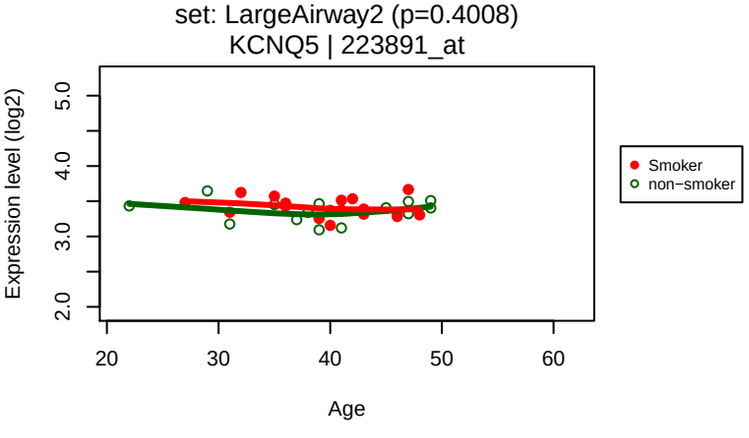

Gene expression ratio (smokers/nonsmokers)
gene: KCNQ5 | 223891_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.798
p=0.6854
p=0.5511
p=0.0746
p=0.4008
p=0.8655
p=0.3465
p=0.1171
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

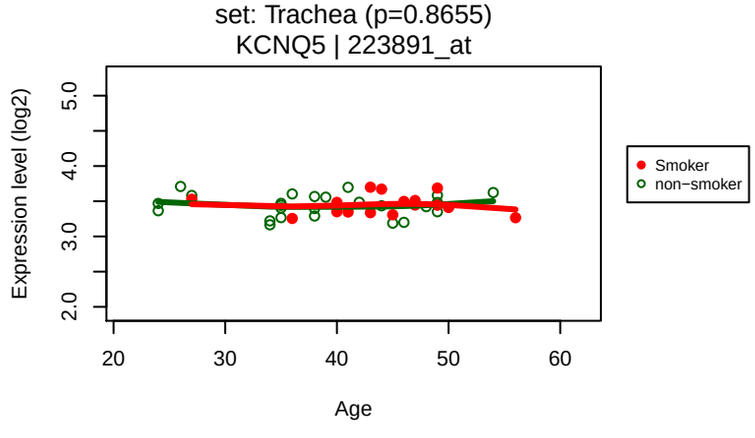

Gene expression ratio in Males
gene: KCNQ5 | 223891_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7968
p=0.9316
p=0.6905
p=0.5428
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KCNQ5 | 223891_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.558
p=0.2566
p=0.9174
p=0.0389
SmallAirway2
LargeAirway2
Trachea
Macrophage2

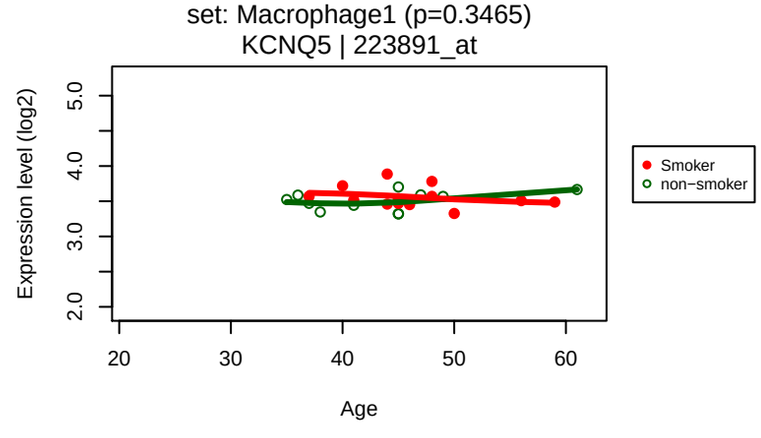

set: SmallAirway1 (p=0.798)
KCNQ5 | 223891_at
Expression level (log2)
Age
Smoker
non−smoker

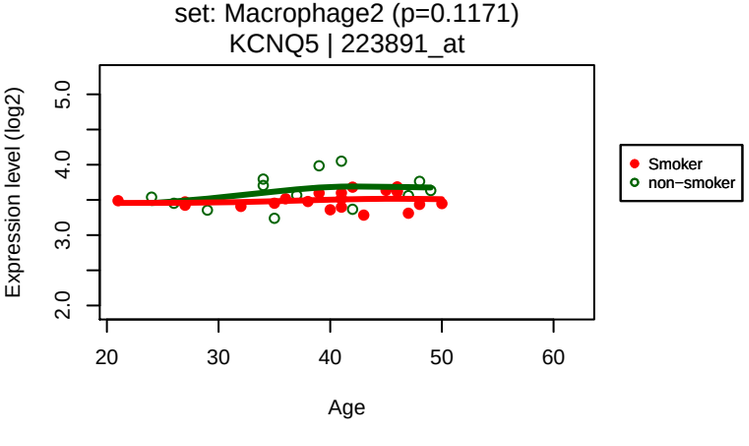

set: SmallAirway2 (p=0.6854)
KCNQ5 | 223891_at
Expression level (log2)
Age
Smoker
non−smoker

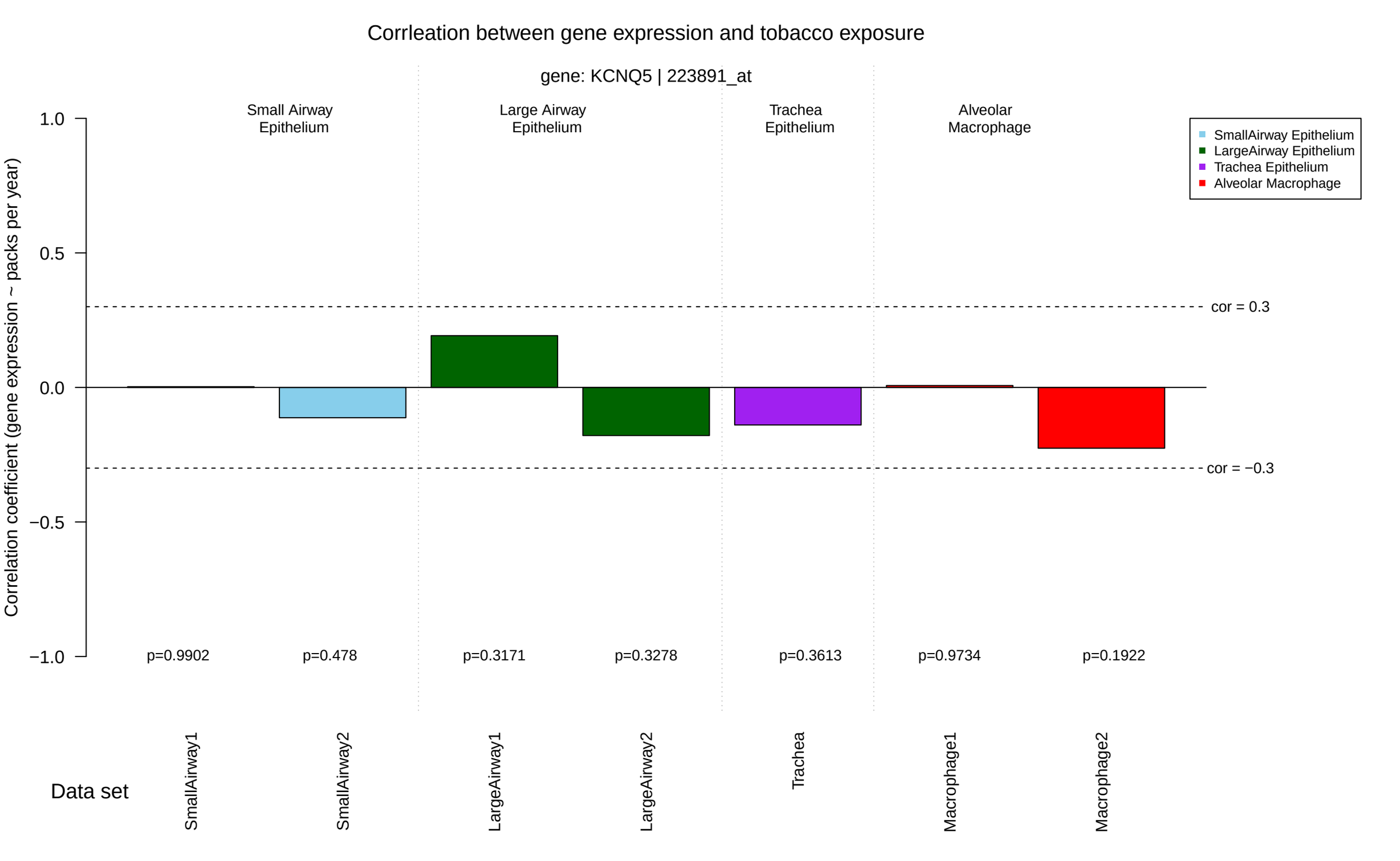

set: LargeAirway1 (p=0.0746)
KCNQ5 | 223891_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4008)
KCNQ5 | 223891_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8655)
KCNQ5 | 223891_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3465)
KCNQ5 | 223891_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1171)
KCNQ5 | 223891_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KCNQ5 | 223891_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9902
p=0.478
p=0.3171
p=0.3278
p=0.3613
p=0.9734
p=0.1922
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2